

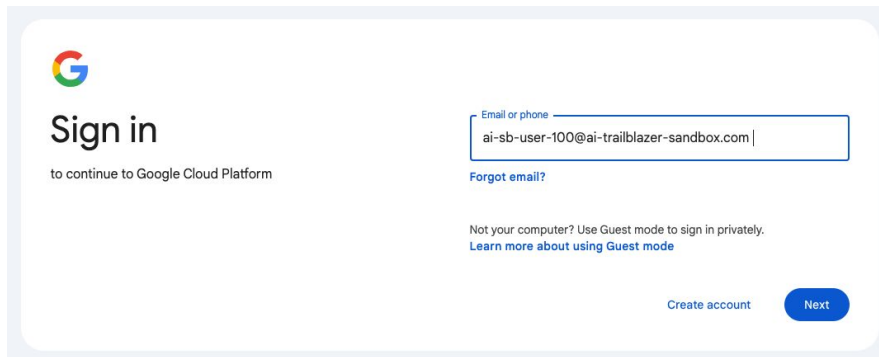
LAB GUIDE

AI Labs with Google AI

Labs Access: Accessing Google Cloud Console

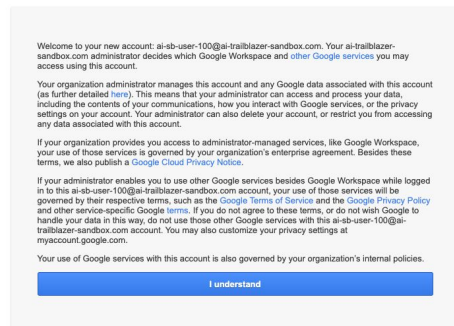
Step 1: Navigate to Google Cloud Console.

<https://console.cloud.google.com/>



The screenshot shows the Google Cloud Platform sign-in interface. At the top is the Google logo. Below it is the text "Sign in" followed by "to continue to Google Cloud Platform". There is a text input field for "Email or phone" containing the email address "ai-sb-user-100@ai-trailblazer-sandbox.com". To the right of the input field are links for "Forgot email?" and "Not your computer? Use Guest mode to sign in privately. [Learn more about using Guest mode](#)". At the bottom right are two buttons: "Create account" and "Next".

Step 2: For the first login, you have to click “I Understand”



The screenshot shows the "Welcome to your new account" page. It features the Google logo and the heading "Welcome to your new account". Below this is a paragraph of text: "Welcome to your new account: ai-sb-user-100@ai-trailblazer-sandbox.com. Your ai-trailblazer-sandbox.com administrator decides which Google Workspace and other Google services you may access using this account." This is followed by a paragraph: "Your organization administrator manages this account and any Google data associated with this account (as further detailed [here](#)). This means that your administrator can access and process your data, including the contents of your communications, how you interact with Google services, or the privacy settings on your account. Your administrator can also delete your account, or restrict you from accessing any data associated with this account." Then another paragraph: "If your organization provides you access to administrator-managed services, like Google Workspace, your use of those services is governed by your organization's enterprise agreement. Besides these terms, we also publish a [Google Cloud Privacy Notice](#)." This is followed by a paragraph: "If your administrator enables you to use other Google services besides Google Workspace while logged in to this ai-sb-user-100@ai-trailblazer-sandbox.com account, your use of those services will be governed by their respective terms, such as the [Google Terms of Service](#) and the [Google Privacy Policy](#) and other service-specific Google terms. If you do not agree to these terms, or do not wish Google to handle your data in this way, do not use those other Google services with this ai-sb-user-100@ai-trailblazer-sandbox.com account. You may also customize your privacy settings at [myaccount.google.com](#)." Finally, a paragraph: "Your use of Google services with this account is also governed by your organization's internal policies." At the bottom is a blue button labeled "I understand".

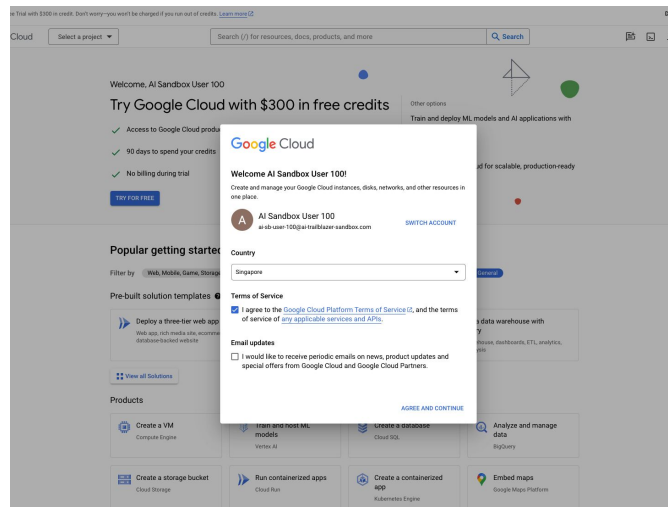
Labs Access: Accessing Google Cloud Console

Proprietary + Confidential

Step 3: You will be prompted to change password. Please enter a new password.



Step 4: Check the checkbox for “Terms of Service” and then proceed to click on “Agree and Continue”.



Labs Access: Accessing Google Cloud Console

Proprietary + Confidential

Navigate to Vertex AI and then Workbench

The screenshot shows the Google Cloud console interface. In the left-hand navigation menu, the 'Vertex AI' option is highlighted with a red box. Below it, under the 'NOTEBOOKS' section, the 'Workbench' option is also highlighted with a red box. The main content area displays the 'Workbench' page, which includes tabs for 'INSTANCES', 'EXECUTIONS', and 'SCHEDULES'. The 'INSTANCES' tab is active, showing a list of Workbench instances. The table lists three instances, each with a status icon, instance name, a link to 'OPEN JUPYTERLAB', zone, auto upgrade status, version, machine type, GPUs, and owner.

Instance name	Zone	Auto upgrade	Version	Machine Type	GPUs	Owner
ai-sb-user-523-gpu	asia-southeast1-a	—	M124	Graphics Optimized: 1 NVIDIA L4 GPU, 4 vCPUs, 16GB RAM	NVIDIA L4 x 1	391264778563-compute@developer.gserviceaccount.c
ai-sb-user-523-no-gpu	asia-southeast1-a	—	M124	Efficient Instance: 4 vCPUs, 16 GB RAM	None	391264778563-compute@developer.gserviceaccount.c
ai-sb-user-524-gpu	asia-southeast1-a	—	M124	Graphics Optimized: 1 NVIDIA L4 GPU, 4 vCPUs, 16GB RAM	NVIDIA L4 x 1	391264778563-compute@developer.gserviceaccount.c

Labs Setup: Accessing Workbench Notebook

Proprietary + Confidential

Step 4: Once at Workbench, ensure you are on the **INSTANCES** tab

Step 5: You should see that the notebooks created for your team.
Each user is assigned 2 notebooks. 1 with GPU and another without GPU.

instance: ai-no-gpu-100--XXX

Google Cloud | ai-sandbox-Company17 | Search (/) for resources, docs, products, and more | Search

Vertex AI | Workbench | CREATE NEW | REFRESH | LEARN

TOOLS: Dashboard, Model Garden, Pipelines

NOTEBOOKS: Colab Enterprise, Workbench

VERTEX AI STUDIO: Overview, Multimodal, Language, Vision, Speech

DATA

View: **INSTANCES** | USER-MANAGED NOTEBOOKS | MANAGED NOTEBOOKS

Workbench Instances have JupyterLab 3 pre-installed and are configured with GPU-enabled machine learning frameworks. [Learn more](#)

Filter

	Instance name	Zone	Auto upgrade	Version	Machine Type	GPUs	Owner	Created	Labels
☒	ai-gpu-100-3ffb19a7 OPEN JUPYTERLAB	asia-southeast1-c	—	M118	Graphics Optimized: 1 NVIDIA L4 GPU, 4 vCPUs, 16GB RAM	NVIDIA L4 x 1	55282892569-compute@developer.gserviceaccount.com	Mar 11, 2024, 7:49:27 AM	consumer.p...: ai-sandbox...
☒	ai-sb-user-100-no-gpu OPEN JUPYTERLAB	asia-southeast1-c	—	M118	Efficient Instance: 4 vCPUs, 16 GB RAM	None	55282892569-compute@developer.gserviceaccount.com	Mar 11, 2024, 7:56:19 AM	consumer.p...: ai-sandbox...

Labs Setup: Accessing Workbench Notebook

Step 6: Select the 2 instances assigned and click on **START** in the top menu. If the instances are already started, skip to step 8.

The screenshot shows the Google Cloud Vertex AI Workbench interface. At the top, there's a header with the Google Cloud logo, a dropdown menu for 'ai-sandbox-Company17', a search bar, and buttons for 'DISMISS' and 'START FREE'. Below the header, a blue banner indicates '2 instances selected' with a red box highlighting the 'START' button, along with 'RESET', 'STOP', and 'DELETE' buttons. The left sidebar contains navigation options: TOOLS (Dashboard, Model Garden, Pipelines), NOTEBOOKS (Colab Enterprise, Workbench), and VERTEX AI STUDIO (Overview, Multimodal, Language, Vision, Speech). The main content area shows a table of instances with columns for Instance name, Zone, Auto upgrade, Version, Machine Type, GPUs, Owner, Created, and Labels. Two instances are listed, both with checkboxes selected. The first instance is 'ai-sb-user-100-gpu' and the second is 'ai-sb-user-100-no-gpu'. Both are in the 'asia-southeast1-c' zone, version 'M118', and created on Mar 11, 2024.

Instance name	Zone	Auto upgrade	Version	Machine Type	GPUs	Owner	Created	Labels
☒ ai-sb-user-100-gpu	asia-southeast1-c	—	M118	Graphics Optimized: 1 NVIDIA L4 GPU, 4 vCPUs, 16GB RAM	NVIDIA L4 x 1	55282892569-compute@developer.gserviceaccount.com	Mar 11, 2024, 7:49:27 AM	consumer-p...: ai-sandbox...
☒ ai-sb-user-100-no-gpu	asia-southeast1-c	—	M118	Efficient Instance: 4 vCPUs, 16 GB RAM	None	55282892569-compute@developer.gserviceaccount.com	Mar 11, 2024, 7:56:19 AM	consumer-p...: ai-sandbox...

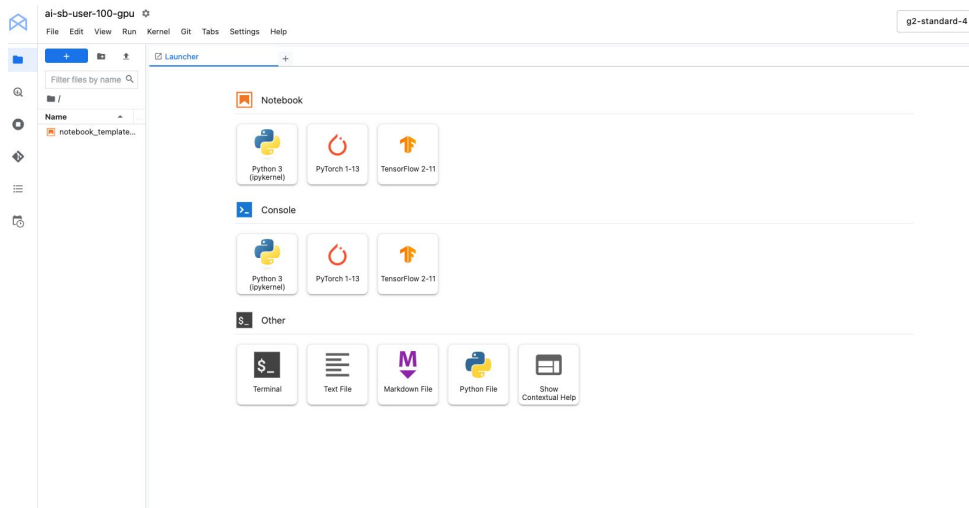
Labs Setup: Accessing Workbench Notebook

Step 7: Wait 1-3 minutes for the instances to start. Verify that there is a green tick beside the instances.



Instance name	Zone	Auto upgrade	Version	Machine Type	GPUs	Owner
ai-sb-user-100-gpu	asia-southeast1-c	—	M118	Graphics Optimized: 1 NVIDIA L4 GPU, 4 vCPUs, 16GB RAM	NVIDIA L4 x 1	55282892569- compute@developer.gserviceaccount.com

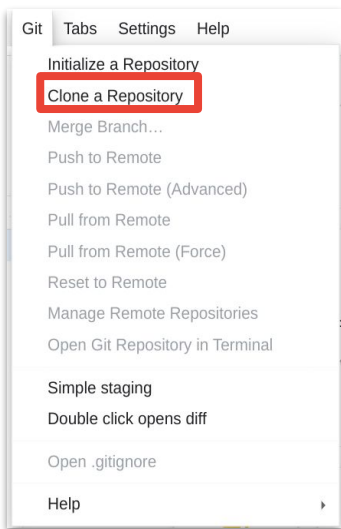
Step 8: Click on **OPEN JUPYTERLAB** for the **gpu instance** <ai-gpu-100-XXX>. A new tab will open with access to the Jupyter Notebook.



Labs Setup: Accessing Workbench Notebook

Step 9: Expand the **Git** menu and click on **Clone a Repository**

Step 10: Paste the URL provided for you into the text box and click **Clone**
<https://github.com/analyticsrepo01/ai-workshop-my>



Lab :

Go to this github URL and clone

<https://github.com/analyticsrepo01/ai-workshop-my/tree/main>

`git clone <URL>`



RAG notebook



Labs Setup: Setting Up Notebooks

Step 11: Double click [RAG_VertexSearch.ipynb](#) to open the next notebook

- my_video.avi
-  part_DIY_RAG_pdf.ipynb
-  part_mixtral.ipynb
-  part_stable_diffusion_2_1.ipynb
-  **RAG_VertexSearch.ipynb**
-  part6_Gemin_on_video.ipynb

Labs Setup: Setting Up Notebooks

Step 12: Expand the **Run** menu and select **Run All Cells**.

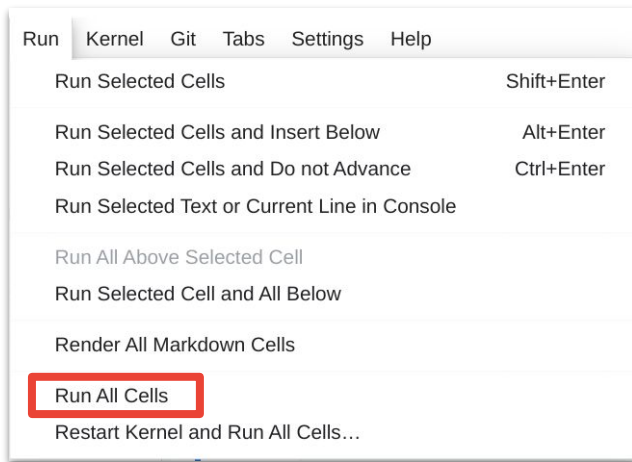


Imagen - to generate & customize images



Labs Setup: Tuning a Open source LLM

Step 13: On Google Cloud Console, Navigate to Vertex AI → Model Garden

	Blog_post.md	2 days ago
	Research.md	2 days ago
	multimodal_use_case.ip...	2 days ago
	part_Imagen.ipynb	3 days ago
	questions_test.json	3 days ago

Gemini for Multimodels usecase

`multimodal_use_case.ipynb`

Labs Setup: Tuning a Open source LLM

Step 13: On Google Cloud Console, Navigate to Vertex AI → Model Garden

	Blog_post.md	2 days ago
	Research.md	2 days ago
	multimodal_use_case.ip...	2 days ago
	part_Imagen.ipynb	3 days ago
	questions_test.json	3 days ago

Agents

New Launcher (🔍 L) +

Filter files by name 🔍

/ GenAI9 / ai-workshop /

Name	Last Modified
campaign	3 days ago
vertex-ai-samples	3 days ago
images1	3 days ago
testing	3 days ago
dog	3 days ago
video	3 days ago
images	3 days ago
books	3 days ago
results	3 days ago
Agent_Example.ipynb	9 minutes ago
README.md	11 minutes ago
RAG_VertexSearch.i...	16 minutes ago
~read	2 days ago
Marketing_post.md	2 days ago
Blog_post.md	2 days ago
Research.md	2 days ago
multimodal_use_ca...	2 days ago
part_Imagen.ipynb	3 days ago
questions_test.json	3 days ago
harry_potter_qa_m...	3 days ago
part_Gemini_on_vid...	3 days ago
harry_potter_RAG_...	3 days ago
questions.json	3 days ago
requirements.txt	3 days ago
harry_potte_qa.csv	3 days ago

Terminal × Agent_E × Terminal × Gemma3 × evnexa × Manage_ × sft.ipynb × slide_ger × AgentEnç × querying_ × +

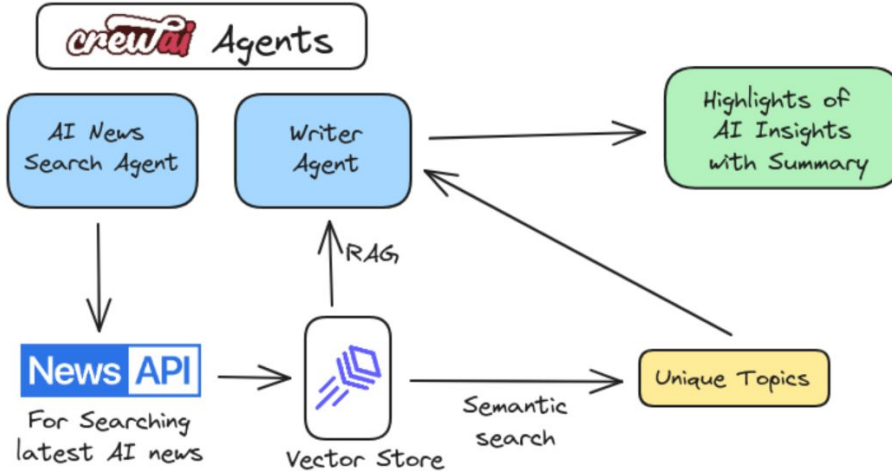
📄 + ✂️ 📄 ▶️ ⏸️ ⏪ ⏩ Code ⌵ ⌚ git ⌚ Execute py312 (Local) 📅

[]: 1

AI Trends Searcher using CrewAI RAG

Using Crew AI to create sophisticated AI news search and writer agents. Using CrewAI RAG create AI Assistants to Run News Agency for AI trends with summarized reports.

This innovative approach combines Retrieving and Generating information, revolutionising how we search for AI news articles, analyse them, and deliver in-depth reports.



Get Free News API Key at [link](#)

Install required packages

Lab :

Go to this github URL and clone

<https://github.com/analyticsrepo01/ai-workshop-my/tree/main>

git clone <URL>



Thank you

